

# Family-Conditioned Multiview Fitting for Semantic Primitive Proxy Actors in UAV Digital Twins (SPPA-MVFit)

Authors omitted for review

## Abstract

Object-level UAV telemetry is not a mesh: a detector may provide a class family, confidence, track identifier, bounding region, and approximate pose, but no occupancy geometry. This paper studies whether a frozen semantic-family part graph improves lightweight 3D occupancy relative to an input-matched, equal-budget nonsemantic graph when both consume the same calibrated top and side silhouettes. The method, family-conditioned SPPA-MVFit, adjusts five shared parameters of a fixed eight-part primitive actor with a 31-candidate silhouette objective; text-only and silhouette-fitted modes call the same actor builder.

On a preregistered developer-held-out synthetic test of 240 actors balanced across six morphology families and two source strata (CSG-ID and structurally different implicit-OOD), the primary paired mean voxel-IoU difference of SPPA-MVFit minus Generic-MVFit is 0.190 with stratified bootstrap 95% CI [0.181, 0.199]. The lower bound exceeds the prespecified +0.030 superiority margin, so the primary hypothesis passes. Both strata remain positive (0.209 CSG-ID; 0.172 implicit-OOD). Secondary comparisons favour SPPA-MVFit over text-only SPPA and lightweight geometric baselines; median clean-call wall time is 9.4 ms (p95 10.6 ms) on the benchmark machine. The result is synthetic internal evidence only: it does not establish real UAV reconstruction, measured flight validity, operator benefit, or an image-to-3D SOTA ranking. A separate SPPA descriptor/update contract and optional Unreal Engine backend are reported as deployment context, not as the primary scientific endpoint.

**Keywords:** semantic 3D proxy; multiview silhouette fitting; UAV digital twin; family-conditioned part graph; primitive actor; occupancy IoU.

## 1 Introduction

Semantic-telemetry UAV digital twins replace part of the continuous video stream with object-level messages. A UAV-side perception stack can report object tracks, classes, confidence, positions, and approximate geometry. The ground station then has to represent those objects in an Unreal/Cesium digital twin. That architectural choice creates a rendering problem that is separate from the flight system itself: object-level telemetry is not a mesh.

Curated 3D assets are useful when the class is known, the asset library is complete, and run-time loading is acceptable. Those assumptions are weak for field UAV operation. Long-tail labels, ambiguous detections, partial views, occlusion, motion blur, and changing object proportions make a fixed asset library brittle. Neural image-to-3D and text-to-3D methods address a different objective: visual asset generation. They can produce richer geometry, but they depend on usable crops or prompts, GPU inference, and mesh outputs that may be too dense for a portable flight laptop that is also running Unreal, telemetry, video, and tracking under a shared rendering budget.

SPPA proposes a lower-fidelity representation point. Instead of reconstructing the real mesh, it compiles a detection into a semantic part graph made from primitive boxes, cylinders, ellipsoids, cones, and tori. A vehicle becomes a cab, body/cargo region, wheels, and optional top details. A quadruped becomes a torso, head, and legs. A tree becomes a trunk and canopy volumes. Unknown or unsupported labels are not hallucinated into specific objects; they fall back to a generic uncertainty-aware display volume with explicit uncertainty metadata.

The scientific question is therefore not whether SPPA looks better than neural 3D generation. It should not. The measurable question is whether a frozen semantic-family part graph improves lightweight 3D occupancy relative to an input-matched, equal-budget nonsemantic graph when both consume the same calibrated silhouettes. Runtime persistence and updateability are reported as separate engineering properties rather than collapsed into a utility score.

## 2 Contribution and Scope

The scientific contribution is **family-conditioned SPPA-MVFit**: evidence that a frozen semantic-family part graph improves 3D occupancy over an equal-budget nonsemantic eight-part graph under identical calibrated top/side silhouettes, parameter count, objective, bounds, initialization, and candidate budget. The knowledge representation under test is the family graph; the optimizer is deliberately simple and is not claimed as a novel fitting algorithm. The method is family-conditioned, not class-agnostic open-set fitting, and not universal image-to-3D reconstruction.

Secondary contributions provide deployment context only:

- **Shared generator contract.** Text-only and silhouette-fitted SPPA modes call the same actor builder and part graph; silhouette evidence changes only the five-parameter vector  $\theta$ .
- **Runtime descriptor/update layer.** SPPA-DESC-0.2/ SPPA-UPD-0.2 separate create from pose/material/shape updates for optional Unreal consumption alongside curated assets.
- **Evidence boundary.** Confirmatory occupancy evidence is developer-held-out synthetic geometry. Real images in this package are qualitative probes without 3D ground truth. Operator benefit, live flight, dense VR readiness, and SOTA image-to-3D ranking remain outside the claim.

### SPPA: language-assisted part decomposition compiled into deterministic 3D primitives

Language can draft the recipe offline; runtime compiles a reviewed versioned recipe plus telemetry.

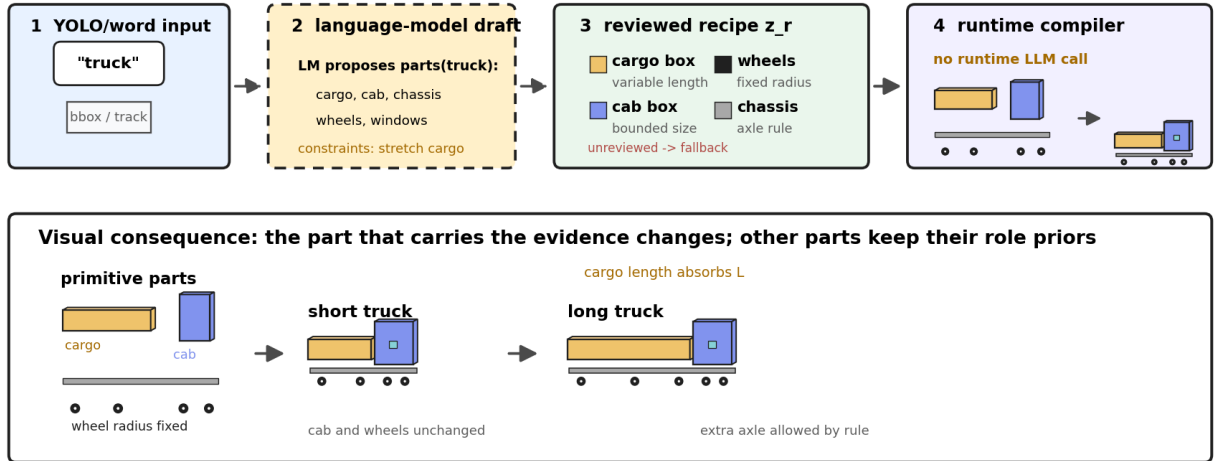


Figure 1: SPPA semantic-label-to-parts-to-3D mechanism for the truck archetype. A language model, ontology tool, or human author may draft candidate part roles offline, but runtime execution uses only a reviewed versioned recipe  $z_r$ . The figure shows the proposed chain explicitly: an input label or word activates an offline draft, the draft is reduced to reviewed part roles and bounded update rules, and runtime compiles those roles into primitive geometry. The short and long proxy variants illustrate the implemented role-specific rule: cargo length changes while cab and wheel priors remain bounded, so the object is not represented by root-scaling a single box.

Claim boundary: Figure 1 is not a claim of photorealistic reconstruction or universally correct geometry; SPPA is only claimed here as a bounded, reviewed semantic-proxy compiler.

Figure 1 states the central representation claim visually: SPPA is a reviewed semantic recipe compiler, not arbitrary word-to-3D generation and not a primitive-shape benchmark. The language component is limited to offline decomposition drafting and ontology expansion. The implemented runtime path starts after an admitted recipe exists: a detector label and telemetry state select a reviewed archetype, fill bounded part parameters, and expose explicit fallback behavior when the label is unsupported. For vehicles, the important distinction is part-parametric adaptation rather than global mesh or root-scale deformation.

### 3 Language Model Use and Geometry Path

The current SPPA implementation does not call a language model in the runtime path. The implemented resolver contract records `resolver_source` as a static keyword ontology and marks `runtime_llm_used` as false; the versioned recipe manifest also disallows runtime LLM use. The executable system described in this paper should therefore not be read as a live LLM that answers questions such as “is this a mammal?” or “how many legs should it have?” before constructing a mesh. Those facts are already encoded in reviewed archetype recipes. If an LLM is used at all, it is only an offline authoring aid that can propose candidate part decompositions; those candidates must be reviewed, versioned, regression tested, and admitted before runtime use. The present artifacts record that offline assistance category, but they do not record a specific LLM provider, model name, prompt, or output hash, so the paper makes no claim about a particular language model.

Two input modes are therefore distinct. A detector-driven input contains a class label plus confidence, track state, bounding region, optional mask, pose, and possibly calibrated scale. SPPA may use that evidence to choose a family, derive an axial footprint, update pose, or adapt dimensions. A tag-only input contains only a word or phrase; it can select a reviewed recipe and default dimension priors, but it cannot provide real silhouette, pose, color, or metric scale evidence. This distinction is important for the benchmark figures: a text-only prompt activates a recipe, whereas a detected object can carry measurement evidence.

Runtime execution is a deterministic compilation pipeline:

1. A detector label or tag is normalized to a SPPA tag and runtime label. For example, labels such as `electric pylon`, `agricultural vehicle`, or `vehicle` are mapped to reviewed semantic families such as vertical structure, farm vehicle, heavy vehicle, or generic vehicle when the evidence supports only that level.
2. The generator resolves the normalized label through exact-class rules, keyword-archetype rules, verifier-gated open-label proxy recipes, or explicit unknown fallback.
3. Metric dimensions come from supplied dimensions, calibrated footprint evidence, or bounded archetype priors. Without calibration, a mask remains image-space evidence and does not by itself become metric geometry.
4. The selected recipe emits role-labeled primitives, such as body, cab, cargo, wheel, torso, head, leg, trunk, canopy, mast, crossbar, or fallback volume. When parts are semantically attached, the recipe also emits explicit lightweight connector cylinders between contact points, for example bicycle frame tubes, rider-to-bike limbs, tower braces, or tractor-trailer hitches. Each primitive records type, local transform, scale, material role, and uncertainty metadata.
5. The descriptor is consumed by Unreal, which creates primitive components once per track and archetype and then applies pose, confidence, material, or compatible shape updates without rerunning a generative model.

For an animal-like object, SPPA does not infer anatomy from first principles or edit a mesh by hand at runtime. A label such as **cow**, **horse**, **dog**, or a conservative **quadruped** family selects a semantic morphology profile inside the same quadruped generator. Those profiles control torso proportions, head/neck placement, legs, tail, horn/h hoof hints, and optional markings, while the emitted proxy remains a low-polygon part graph rather than a watertight biological reconstruction. This explains why **biker** is visually cleaner from text alone: wheels and frame tubes are discrete geometric roles, whereas an animal is an organic shape represented only by bounded semantic priors unless image or metric evidence is available. A local text-only animal probe over **cow**, **bull**, **horse**, **dog**, **sheep**, **goat**, **deer**, and **quadruped** generated all cases through this shared profile path, with 836–1180 triangles and at most 0.61 ms in-process part synthesis before export. For a tractor or tractor-trailer, the system need not detect the exact taxonomic class to be useful: a detector output such as **vehicle**, **agricultural vehicle**, **truck**, or **trailer** can be normalized to a farm, heavy, or generic vehicle proxy. If the label is weak or adversarial, SPPA should prefer a conservative fallback volume over inventing a specific object. In the current implementation, a small set of useful UAV open-label cases such as **forklift**, **traffic cone**, **water tank**, **shipping container**, **drone**, and **hay bale** may pass through a deterministic recipe verifier before they are accepted as generated proxy recipes; otherwise fallback remains the runtime behavior.

This is also the honest novelty boundary. Primitive composition, CSG, shape programs, procedural modeling, silhouette consistency, multiview primitive optimization, and text-to-CAD are mature research areas [2, 43, 51, 64, 42, 32, 18, 19, 21, 40, 50, 12, 30]. SPPA’s contribution is not that boxes and cylinders can describe objects. Its contribution is a bounded UAV-runtime contract that converts imperfect semantic evidence into lightweight, controllable, updateable, and fallback-safe 3D proxy actors.

## 4 Representation Design Space

The relevant comparison is not a single “3D representation” baseline. A UAV digital twin can visualize a semantic detection through several families of representations, each making different assumptions about input evidence, runtime budget, geometry, and operator interpretation. Geometry-only displays such as boxes, ellipsoids, or billboards remain useful lower-bound controls, but they are not the scientific competitor. SPPA targets the gap between a generic box and a curated asset: a role-labeled primitive actor when only sparse semantic telemetry is available.

Table 1 states the resulting deployment policy. It deliberately gives priority to curated assets and mapping geometry when those are the correct tools.

Table 1: Operational representation policy for semantic detections.

| Situation                       | Preferred representation       | Reason                                                          | Boundary                                 |
|---------------------------------|--------------------------------|-----------------------------------------------------------------|------------------------------------------|
| Matched project asset exists    | Curated asset backend          | best appearance and lower measured cost in the cow/tower subset | asset library must match class and scale |
| Known semantic family, no asset | SPPA primitive actor           | persistent, local, role-labeled 3D proxy from label/pose input  | finite reviewed archetypes               |
| Unknown or adversarial label    | SPPA fallback volume           | avoids inventing a specific object from weak semantics          | not visually specific                    |
| Clean crop plus offline budget  | Image-to-3D or asset authoring | richer mesh can be generated outside the flight update loop     | no per-frame track contract              |
| Planning or occupancy task      | Occupancy / ESDF / map layer   | represents navigable space and collision evidence               | not a role-labeled display actor         |

## 5 Related Work

SPPA overlaps with several mature research areas, but it does not replace them. Text-to-3D and image-to-3D generation methods such as DreamFusion, Magic3D, Point-E, Shap-E, TripoSR, TripoSG/Tripo P1, InstantMesh, Stable Fast 3D, SPAR3D, LGM, CRM, Unique3D, Direct3D-S2, PartCrafter, Pixa3D, TRELLIS/TRELLIS.2, Hunyuan3D 2.x, and commercial systems such as Rodin Gen-2.5 target visual asset generation or reconstruction [37, 26, 33, 20, 49, 24, 59, 44, 3, 45, 46, 53, 54, 55, 27, 23, 58, 57, 63, 47, 17]. Their strength is fidelity when suitable visual or prompt evidence is available. SPPA targets a different runtime point: deterministic actor generation from sparse semantic telemetry, low polygon counts, and cheap per-frame updates. Recent benchmark work also highlights why a real image-to-3D ranking needs a shared reference set, metrics, and part/structure evaluation rather than a local figure assembled from heterogeneous inputs [48, 60]. For this reason, the artifact package includes an explicit SOTA-protocol readiness manifest and verifier. In the current artifact state that verifier keeps the comparative figure classified as an input-alignment audit, not as an image-to-3D SOTA leaderboard.

Monocular and open-vocabulary 3D detection estimate 3D boxes or object state from images [4, 61]; YOLOE supplies approximate open-prompt detection/segmentation evidence rather than universal recognition [52]. Animal pose and animal reconstruction methods estimate detailed skeletons or category-specific 3D shape [62, 25, 16]. These methods can provide stronger evidence to SPPA, but SPPA itself is a rendering/backend layer, not a detector or pose estimator.

Primitive representations have a long history. Marr’s volumetric primitives, recognition-by-components, superquadrics, learned part decomposition, recent superquadric scene decomposition, residual primitive fitting, and positive/negative multiview primitives show that structured primitive geometry can encode compact shape abstractions [29, 2, 43, 36, 51, 31, 11, 12, 30]. Neural primitive and CSG parsers show that networks can infer compact primitive or boolean programs from shape evidence [64, 42]. Procedural modeling, shape grammars, and learned shape programs show how compact programs can generate or abstract structured objects [32, 18, 19]. Text-to-CAD and point-cloud-to-code systems extend this program-generation line to natural-language CAD prompts and LLM-assisted CAD reconstruction [21, 40]. SPPA uses these ideas operationally, but the claimed novelty is the bounded, track-updateable semantic telemetry contract inside an Unreal UAV digital twin.

Recent scene-level systems narrow the gap from other directions. SAM 3D decomposes a single image into composable per-object geometry with layout [41], and PartCrafter generates part-structured meshes from one image [27]; both are heavy offline reconstruction or generation pipelines, not millisecond telemetry-driven proxies with an update contract. SuperQuadricOcc reaches real-time semantic occupancy with superquadric volume rendering in driving scenes [14], and lightweight urban twins render proxy building geometry at interactive rates [28]; neither constructs role-labeled, metric-scaled object actors from detector tags and sparse silhouettes. Test-time spatial control steers generative 3D models with coarse primitives [10], which complements rather than replaces a deterministic proxy contract.

Graphics systems already provide level-of-detail, impostors, billboards, streaming tiles, and virtualized geometry [5, 6, 9, 35]. 3D and dynamic scene graphs maintain semantic world structure [1, 39, 56]. Occupancy grids, OctoMap, ESDF, and TSDF maps support planning and mapping [15, 34]. SPPA should be read as a display actor representation that can consume such upstream evidence, not as a mapping substrate.

Finally, any operator-facing claim must be evaluated with human-factors methods. SAGAT, situation-awareness theory, and NASA-TLX are relevant to a future SPPA operator study [7, 8, 13]. The present paper contains no pilot study.

## 6 SPPA Contract

At time  $t$ , SPPA consumes an object state

$$x_t = \{i, \ell, c, b_t, m_t, \xi_t, s_t, \Sigma_t, h_t\},$$

where  $i$  is a track identifier,  $\ell$  a detector label,  $c$  confidence,  $b_t$  a bounding region,  $m_t$  an optional mask or footprint,  $\xi_t$  a world pose estimate,  $s_t$  optional metric dimensions,  $\Sigma_t$  uncertainty, and  $h_t$  the track history. The resolver maps  $\ell$  to one of three cases:

- **exact\_class**: implemented label with a fixed archetype;
- **keyword\_archetype**: out-of-template label mapped to a finite reviewed archetype, such as vehicle or vertical structure;
- **fallback\_unknown\_label**: unknown label rendered as a generic uncertainty-aware volume.

This is deliberately not universal word-to-3D generation. SPPA can accept an open detector tag at runtime, but the tag is normalized into a reviewed family or a conservative unknown fallback before geometry is emitted. A language model may assist offline ontology expansion, but runtime resolution must remain bounded, reviewed, cached, and able to reject over-specific mappings. Figure 1 makes this boundary explicit: language-model support belongs to offline archetype authoring or review, while the runtime path maps a detector label to a reviewed archetype, generic fallback volume, descriptor parts, and Unreal components. The SPPA observation path makes the metric-evidence boundary explicit through a **SPPA-OBS-0.1** observation record. It stores detector evidence, camera and flight geometry, a projected local footprint when available, a metric dimension proposal, and a covariance-style uncertainty envelope. Unless the record marks telemetry and metric ground truth as measured, the resulting pose and scale are treated as scenario-relative evidence for a proxy contract rather than as flight-localization truth. The current implementation includes a versioned archetype-recipe contract with 15 reviewed archetypes and 64 exact/keyword/fallback label checks. That contract verifies resolver status, allowed runtime archetype, required semantic material roles, and disabled runtime LLM use; it does not validate geometric correctness for arbitrary open-vocabulary labels. A separate offline recipe-review gate now tests five candidate mappings before caching: one accepted reviewed-family mapping, one fallback-only mapping, and three rejected mappings for visual artifact context, unreviewed topology/material roles, or attempted runtime LLM use. This is an implementation guard, not proof of open-vocabulary coverage.

### 6.1 Part Graph

Each archetype  $a$  defines a part graph

$$G_a = (V_a, E_a), \quad v_k = (p_k, r_k, \tau_k, \rho_k),$$

where  $p_k$  is a primitive type,  $r_k$  is its role,  $\tau_k$  its transform relative to the actor root, and  $\rho_k$  role-specific parameters such as dimension priors, color/material role, and confidence tags. The current implementation uses boxes, cylinders, ellipsoids, cones, and tori.

Vehicle adaptation is the clearest implemented case. A long truck and a short truck should not be produced by globally scaling the same box. In the current vehicle grammar, cab and wheel dimensions remain bounded by role priors, while the cargo/body span and axle placement absorb length changes. In the synthetic truck check, two descriptors with equal width/height and different length keep cab scale fixed at [1.656, 2.070, 1.952] m and wheel scale fixed at [0.392, 0.094] m, while cargo/body length changes from 2.263 m to 5.188 m. This is an internal part-invariance diagnostic, not a visual comparison against competing 3D generation methods and not proof that SPPA recovers true truck morphology from UAV imagery. The corresponding component views are kept in the artifact log rather than used as main-paper comparative evidence.

## 6.2 Pose, Yaw, and Updates

The descriptor separates actor pose  $\xi_t$  from part parameters  $q_t$ . Pose changes are per-frame updates; topology regeneration is avoided unless the class/archetype changes or new evidence crosses a configured threshold. A typical update packet therefore changes only root transform, confidence, material/uncertainty tags, or a small set of shape parameters.

Yaw ambiguity is handled explicitly. A top-down footprint or PCA orientation gives an axial orientation modulo  $\pi$ , not necessarily a signed heading modulo  $2\pi$ . Signed yaw requires velocity, asymmetry, multi-frame evidence, or an external heading source. If no reliable sign exists, the descriptor sets `yaw_ambiguous=true`. The exact thresholds and weights in the fitting objective are exposed as versioned protocol parameters in the current implementation, not calibrated constants.

When a mask polygon is available, the descriptor records both the axis-aligned mask box and an oriented image-space footprint from PCA projections. This is used as footprint evidence and axial yaw evidence. If explicit metric dimensions or an explicit image-to-ground scale are supplied, the current generator can convert the calibrated footprint to metric length/width and pass those dimensions to the part-parametric builder. Without such calibration, the mask remains image-space evidence only and does not change metric geometry.

## 7 Implementation

The prototype has three layers. The first is a Python OBJ/MTL prototype generator for a finite set of explicit and keyword-resolved archetypes, including vehicles, vulnerable road users, vegetation, vertical structures, generic quadrupeds, and unknown fallback volumes. This path is useful for visual inspection and lightweight local timing only.

The second layer emits versioned descriptors and update packets. A `SPPA-DESC-0.2` record stores resolver source, match type, fallback reason, scale source, axis-aligned and oriented mask footprint evidence, yaw ambiguity, shape confidence, material source, optional observed-color evidence, part list, primitive counts, and scheduler metadata. A `SPPA-UPD-0.2` record separates create/regenerate payloads from pose/no-op/shape updates. The design goal is to generate once per track/archetype, update pose/material/uncertainty every frame, and apply shape-parameter changes in place when the part topology is unchanged. When supplied by the SPPA observation bridge, the descriptor additionally preserves the observation identifier, covariance matrix, position/scale/yaw uncertainty estimates, and visual uncertainty policy. These fields are carried as auditable metadata; they do not make the visual uncertainty encoding a validated human-factors result. In the current generator, `descriptor_id` is a topology-cache identity: it excludes pose, timestamp, confidence, and observed color, which remain in `input_hash` or update fields. This separation is required for cheap per-frame updates; otherwise a pose update would look like a new mesh descriptor.

The third layer is an optional Unreal Engine 5.7 telemetry backend. The same obstacle payload can drive either curated asset actors or semantic proxy actors. The asset backend remains the default. The SPPA backend can ingest descriptor JSON, construct primitive components from the `parts` array, apply pose/no-op/shape update packets, and reject invalid JSON or mismatched descriptor identifiers without changing the current actor. This implements a runtime switch without forcing a migration away from curated assets.

The current resolver coverage regression checks 34 direct labels, including reviewed archetypes, conservative fallback contexts, and verifier-gated open-label recipes. A separate recipe-manifest contract checks 23 archetypes and 95 exact or keyword labels against the runtime generator. These checks verify resolver consistency and recipe roles; they are not evidence that the ontology is complete.

## 8 Family-Conditioned SPPA-MVFit

### 8.1 Observation contract and shared actor

Every compared method receives the same case identifier, morphology-family token, metric world box, and two  $96 \times 96$  orthographic binary silhouettes (top and side). No method receives RGB pixels, detector scores, source parameters, ground-truth occupancy, evaluation projections, or real or replay telemetry. The forbidden-input list is frozen in the protocol package.

An actor is an ordered list of eight primitive slots (boxes, cylinders, ellipsoids). Family graphs and a single **generic** graph are stored as static JSON. The only builder used by text-only SPPA, Generic-MVFit, and SPPA-MVFit is

$$A = \text{Build}(g, \theta), \quad \theta = (\log s_x, \log s_y, \log s_z, s_{\text{sec}}, o_{\text{sec}}), \quad (1)$$

with identical bounds for every graph  $g$ . Text-only SPPA uses the family graph at the default  $\theta_0$ . Generic-MVFit and SPPA-MVFit optimize the same five parameters with the same silhouette disagreement objective and a fixed 31-candidate local search budget; only the graph identity differs (**generic** vs family). Baselines include axis-aligned box, ellipsoid, capsule, billboard, and nonsemantic visual hull from the same masks.

### 8.2 Preregistered confirmatory protocol

The primary hypothesis compares clean-mask 64-cubed voxel IoU of SPPA-MVFit against Generic-MVFit. Analysis units are 240 source actors (20 per family in each of two strata). The confirmatory estimand is the equal-stratum mean paired difference. H1 passes only if the lower endpoint of a two-sided 95% stratified actor-bootstrap percentile interval (10,000 resamples, seed 77157) exceeds +0.030. Seeds were derived from a NIST Randomness Beacon pulse after a pre-test freeze and a written triple-role protocol release under Amendment 03. Method outputs were sealed and hashed before private ground-truth evaluation. Development results were used only to implement the frozen package and are not confirmatory.

Source actors come from two strata: CSG-ID unions of solids with family topology, and implicit-ODD occupancy from superelliptic extrusions, taper/twist, toroidal sections, and spline-distance tubes. Source and method modules do not import each other. The held-out set is developer-held-out synthetic geometry, not an externally blinded public UAV benchmark.

## 9 Primary Results

Table 2 reports the sealed confirmatory decision. Table 3 reports clean-condition method means and wall-clock costs. Table 4 reports secondary paired differences of SPPA-MVFit against the remaining baselines using the same stratified bootstrap interval construction.

Table 2: Preregistered primary endpoint on the held-out synthetic test ( $n=240$  actors). Provenance: **synthetic\_geometry**. H1 passes only if the stratified bootstrap 95% lower bound exceeds +0.030.

| Quantity                                               | Value          |
|--------------------------------------------------------|----------------|
| Primary mean $\Delta$ IoU (SPPA-MVFit – Generic-MVFit) | 0.190          |
| Stratified bootstrap 95% CI                            | [0.181, 0.199] |
| Prespecified superiority margin                        | +0.030         |
| H1 decision (lower bound > margin)                     | PASS           |
| Actors (analysis units)                                | 240            |
| CSG-ID stratum mean $\Delta$                           | 0.209          |
| Implicit-ODD stratum mean $\Delta$                     | 0.172          |
| Resolution sensitivity 48/64/80                        | PASS           |



Table 3: Clean-mask method summary on the held-out test ( $n=240$ ). Mean/median are 64-cubed voxel IoU. Timings are single-call wall-clock milliseconds on the benchmark machine and are not Unreal frame times.

| Method           | Mean IoU | Median IoU | Median ms | p95 ms |
|------------------|----------|------------|-----------|--------|
| SPPA-MVFit       | 0.557    | 0.563      | 9.43      | 10.60  |
| Generic-MVFit    | 0.367    | 0.390      | 11.34     | 11.81  |
| SPPA text-only   | 0.427    | 0.441      | 0.77      | 1.81   |
| Visual hull      | 0.522    | 0.567      | 0.22      | 0.24   |
| Ellipsoid        | 0.342    | 0.292      | 0.50      | 0.55   |
| Capsule          | 0.325    | 0.260      | 0.95      | 1.07   |
| Axis-aligned box | 0.248    | 0.190      | 0.23      | 0.27   |
| Billboard        | 0.173    | 0.148      | 0.23      | 0.26   |

Table 4: Paired clean-mask mean IoU differences of SPPA-MVFit minus each comparator (stratified actor bootstrap 95% intervals). Visual hull is a high-complexity geometry reference, not a lightweight runtime actor.

| Comparator         | Mean $\Delta$ IoU | 95% CI low | 95% CI high |
|--------------------|-------------------|------------|-------------|
| Generic-MVFit (H1) | 0.190             | 0.181      | 0.199       |
| SPPA text-only     | 0.130             | 0.116      | 0.144       |
| Visual hull        | 0.036             | 0.027      | 0.044       |
| Ellipsoid          | 0.216             | 0.207      | 0.224       |
| Capsule            | 0.233             | 0.224      | 0.241       |
| Axis-aligned box   | 0.310             | 0.301      | 0.318       |
| Billboard          | 0.384             | 0.375      | 0.393       |

Table 5: Paired clean-mask mean IoU differences of SPPA-MVFit minus Generic-MVFit by morphology family and source stratum. All twelve cells are positive; the smallest gain is the compact-vehicle implicit-OOD cell. The equal-stratum aggregate is the primary estimate of Table 2.

| Family                  | CSG-ID $\Delta$ | Implicit-OOD $\Delta$ | Mean $\Delta$ |
|-------------------------|-----------------|-----------------------|---------------|
| articulated_vehicle     | 0.113           | 0.125                 | 0.119         |
| branching_vertical      | 0.200           | 0.210                 | 0.205         |
| compact_vehicle         | 0.101           | 0.043                 | 0.072         |
| lattice_tower           | 0.175           | 0.163                 | 0.169         |
| quadruped               | 0.205           | 0.189                 | 0.197         |
| rider_cycle             | 0.458           | 0.299                 | 0.379         |
| Equal-stratum aggregate | 0.209           | 0.172                 | 0.190         |

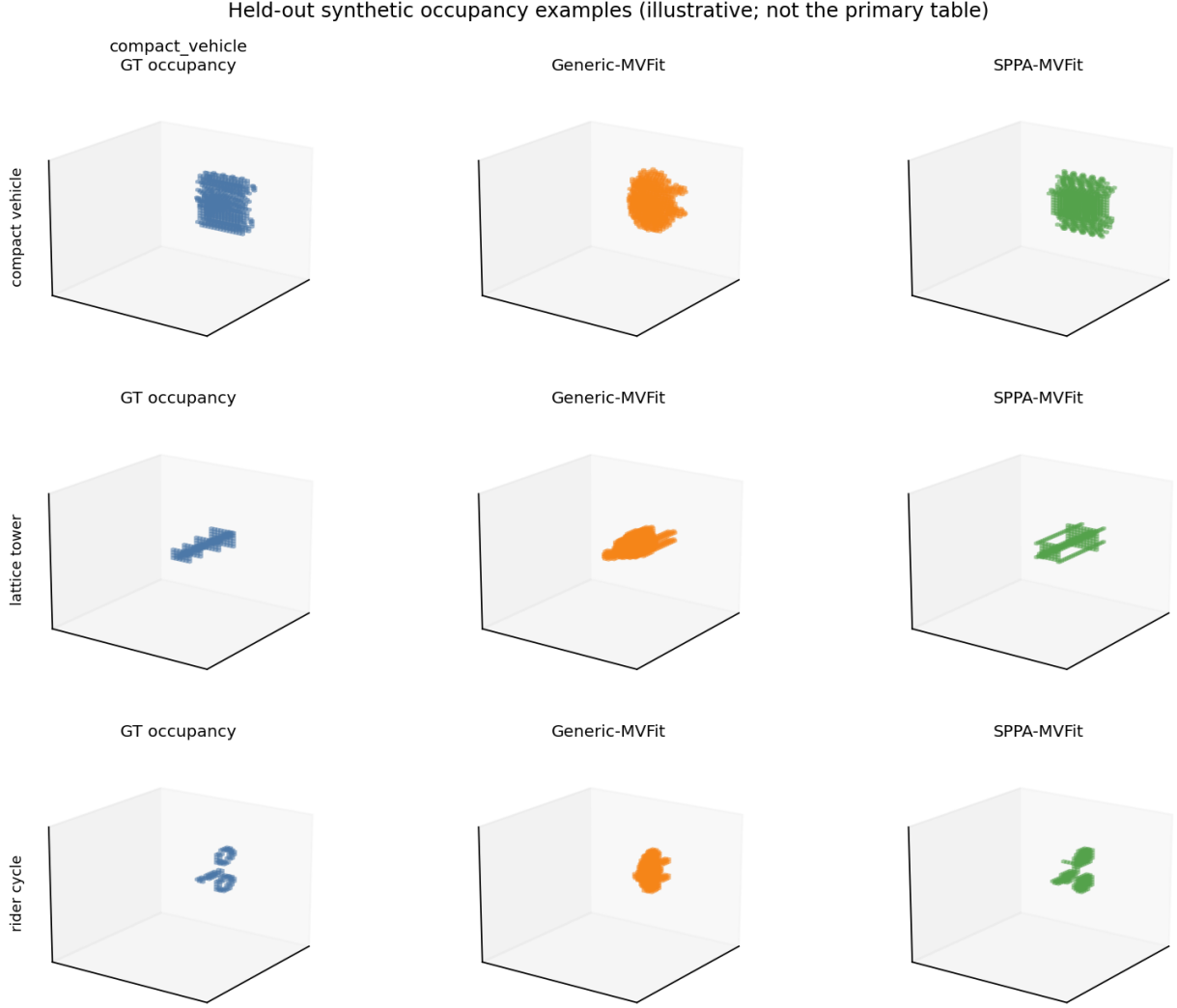


Figure 2: Held-out synthetic occupancy examples for three families: ground truth (left), Generic-MVFit (center), and SPPA-MVFit (right). Illustrative only; the confirmatory decision uses Table 2, not this figure.

H1 passes: mean  $\Delta\text{IoU} = 0.190$  with 95% CI  $[0.181, 0.199]$ . The CSG-ID stratum mean difference is 0.209 and the implicit-OOD stratum mean difference is 0.172; neither is non-positive, so a cross-stratum generalization warning is not triggered by the protocol rule. Table 5 reports the paired difference in every family-by-stratum cell and Figure 2 shows qualitative occupancy examples. SPPA-MVFit also exceeds SPPA text-only (mean  $\Delta = 0.130$ , CI  $[0.117, 0.143]$ ) and all lightweight geometric baselines. Mean IoU of SPPA-MVFit (0.557) is slightly above nonsemantic visual hull [22] (0.522), but visual hull is denser occupancy geometry without a role-labeled lightweight actor contract; the paper therefore does not present visual hull as a defeated runtime competitor. Median SPPA-MVFit time is 9.4 ms (p95 10.6 ms), below the secondary local-cost gates of 100 ms median / 250 ms p95. A resolution-sensitivity check at grids 48/64/80 on the prespecified actor subset passes the protocol threshold.

**Claim boundary:** these occupancy gains are measured on synthetic calibrated silhouettes with known family tokens. They are not UAV flight measurements, not detector accuracy, and not operator-facing benefit.

## 10 Secondary Systems Evidence

The following subsections retain deployment-oriented regressions and qualitative probes. They constrain engineering claims; they do not replace the primary endpoint in Section 9.

### 10.1 Measured External Neural Wave on Sealed Held-Out Cases

The stress test of Table 8 is qualitative and uses ad-hoc local crops. To answer the expected reviewer question—how SPPA-MVFit compares to external neural image-to-3D generators on the *same* sealed held-out cases—a secondary analysis was prospectively registered as protocol Amendment 05 before any measurement. It does not re-seal predictions and does not change H1, its interval, or any sealed artifact.

**Protocol.** Sixty sealed test actors were selected deterministically (five per morphology-by-stratum cell, lexicographic `case_id` order). Each neural generator received two prespecified inputs: (a) a clean-crop shaded oblique render of the source actor, the generator’s best input condition, and (b) the actual  $96\times 96$  top telemetry mask as the input image, the closest visual analog of the channel SPPA consumes. Outputs were aligned to ground truth by a frozen frame convention (a one-time 48-candidate signed-permutation search on twelve disjoint calibration cases, frozen afterwards), uniform scale to the ground-truth bounding box, and ground-truth bounding-box center and yaw frame; no per-case tuning was allowed. Voxel IoU was measured at  $64^3$  with the sealed voxelizer and WORLD frame. Two hard crashes (2/240 runs, Hunyuan3D oblique condition) are reported as exclusions; SF3D (install failure), SPAR3D (gated weights), and TRELLIS.2 (no Windows environment) remain documented environment-level exclusions. SPPA-MVFit, Generic-MVFit, and visual-hull rows are the existing sealed clean-condition results on the same sixty cases.

Table 6: Measured external neural wave (Amendment 05, secondary). Mean voxel IoU at  $64^3$  on the same sixty sealed held-out cases, with parenthesized paired  $\Delta$ IoU versus SPPA-MVFit. Triangles and payload are mesh-output means; generation time is warm per-case wall time (model load excluded) on the same RTX 5090; VRAM is uncapped peak CUDA allocation. SPPA rows are CPU-only and their payload is the update descriptor, not a mesh. Exact per-case values are released in `benchmarks/results/sppa_neural_external_wave.json`.

| Method            | Input          | IoU            | Triangles | ms      | VRAM (MB) | Payload |
|-------------------|----------------|----------------|-----------|---------|-----------|---------|
| SPPA-MVFit (ours) | masks top+side | 0.561          | 536       | 9.2     | –         | 1.45 kB |
| Generic-MVFit     | same masks     | 0.364 (-0.197) | 1,280     | 11.4    | –         | 1.43 kB |
| Visual hull       | same masks     | 0.516 (-0.045) | 51,432    | 0.2     | –         | 32.8 kB |
| TripoSR (a)       | clean render   | 0.128 (-0.433) | 28,145    | 384.1   | 1,869     | 1.47 MB |
| TripoSR (b)       | top mask       | 0.231 (-0.331) | 46,053    | 379.4   | 1,870     | 2.45 MB |
| Hunyuan3D-2m (a)  | clean render   | 0.157 (-0.403) | 686,733   | 1,350.9 | 4,591     | 10.5 MB |
| Hunyuan3D-2m (b)  | top mask       | 0.171 (-0.390) | 3,089,274 | 1,915.5 | 4,747     | 46.5 MB |

**Reading.** On the sealed cases SPPA-MVFit exceeds the measured neural generators on mean voxel IoU by 0.330–0.433 while using  $52\times$ – $5,767\times$  fewer triangle equivalents, running  $41\times$ – $208\times$  faster on CPU only, with zero CUDA memory and a payload  $10^3$ – $10^4\times$  smaller. Two observations sharpen the boundary. First, both generators score *higher* from the harsh top mask than from their generous clean-crop render: single-view image-to-3D models hallucinate occluded structure, whereas a calibrated footprint constrains occupancy directly; telemetry-shaped evidence is not a degraded image but a different measurement channel. Second, the visual hull remains the strongest occupancy baseline per millisecond, but it carries no part roles, no family conditioning, and no update contract. This table measures an operating point, not a leaderboard: condition (a) gives neural generators richer input than telemetry, condition (b) gives them input they were not trained for, photoreal asset quality remains neural territory, and a

visual beauty ranking against SOTA generators remains prohibited by the claim-evidence matrix. Per-case rows, alignment calibration, environment exclusions, and code hashes are released in `benchmarks/results/sppa_neural_external_wave.{json,md,tex}`.

## 10.2 Verifier-Gated Open-Label Probe

We evaluate the open-label route with 20 deliberately awkward labels that were not selected as the original paper demo classes. The set includes heavy equipment, vertical structures, containers, UAV-like objects, agricultural objects, and unsafe visual-artifact contexts. Table 7 shows the progression from the earlier contract-alignment probe to the current verifier-gated recipe implementation. The metric and tag-only conditions use the same labels; metric inputs provide declared dimensions, while tag-only inputs use template priors.

Table 7: Uncached open-label SPPA probe. Family/proxy means non-fallback runtime output. Verified recipes are accepted only after deterministic checks over required roles, fallback-role absence, primitive budget, and dimensions.

| Condition                        | Fallback               | Family/proxy         | Verified recipes | Critical mismatches |
|----------------------------------|------------------------|----------------------|------------------|---------------------|
| Before alignment                 | 12/20 raw, 13/20 norm. | 8/20 raw, 7/20 norm. | 0/20             | 7                   |
| After contract alignment         | 12/20                  | 8/20                 | 0/20             | 0                   |
| Verifier-gated recipes, metric   | 4/20                   | 16/20                | 8/20             | 0                   |
| Verifier-gated recipes, tag-only | 4/20                   | 16/20                | 8/20             | 0                   |

The remaining fallback cases in this probe are intentional: `dog statue`, `toy tractor`, `worker helmet`, and `cow silhouette` are visual-artifact or object-part contexts. This is a stronger claim than pure fallback but still narrower than arbitrary text-to-3D: SPPA accepts a generated proxy only when the recipe contract passes.

## 10.3 OBJ Construction Sanity Check

The original OBJ construction timing covered the initial seven-class prototype set and is retained only as a legacy construction/export sanity check. In that run, per-object export stayed below 0.1 s including Python startup and file export overhead, with meshes ranging from 488 vertices and 396 faces for `biker` to 856 vertices and 786 faces for `truck`. The current generator has since been extended to a broader finite-archetype resolver. These numbers are therefore implementation evidence for the OBJ construction path only, not current-ontology coverage, Unreal instancing timings, or end-to-end detection-to-render latency.

## 10.4 Benchmark Alignment and Generator Substitutability Stress Test

Table 8 is a substitutability stress comparison, not a SOTA ranking. A real SOTA ranking for image-to-3D would require the same detection crops or reference images, preferably with target meshes or structured human-preference scores, and should include contemporary high-fidelity systems such as TRELIS.2, Pixa3D, Direct3D-S2, TripoSG/Tripo P1, Hunyuan3D 2.1, PartCrafter, and Rodin Gen-2.5 alongside fast feed-forward methods such as SF3D, SPAR3D, and TripoSR. SPPA would not be ranked as an image-to-3D asset generator in that protocol, because its intended input is a semantic detection state rather than a visual crop. The local stress test asks a narrower systems question: whether generators that were runnable in the local Windows workstation stack can replace a bounded runtime proxy under a portable-flight profile. Neural generators are expected to produce richer assets when they receive appropriate prompts or images; SPPA’s question is persistence, updateability, and deterministic fallback from sparse telemetry.

All neural runs used the same RTX 5090 workstation with a 6 GB PyTorch CUDA allocator budget. This was an artificial local engineering stress setting, not a physical GPU partition or a measured deployment profile. The cap used PyTorch’s documented memory-fraction mechanism [38]; it does not establish portability to a particular laptop, aircraft computer, or concurrently loaded Unreal runtime.

Table 8: Preliminary July 2026 substitutability stress comparison over six object classes. Times are local wall-clock measurements under constrained input settings, not statistical or end-to-end UAV/VR validation.

| Method                        | Input      | Wall          | Tris              | VRAM     |
|-------------------------------|------------|---------------|-------------------|----------|
| SPPA finite archetype         | label/tag  | 0.0016 s med. | 488–1,668         | 0 MB     |
| Shap-E text $K = 16$          | prompt/tag | 2.0080 s      | 17,884–131,668    | 6,120 MB |
| Point-E text + SDF32          | prompt/tag | 6.4266 s      | 1,026–5,454       | 3,644 MB |
| TripoSR warm 128 <sup>3</sup> | RGBA crop  | 0.4604 s      | 19,284–31,012     | 2,028 MB |
| Hunyuan3D-2mini Turbo         | RGBA crop  | 1.5042 s      | 332,020–1,698,032 | 6,092 MB |

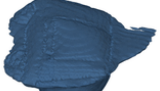
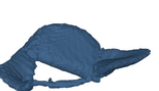
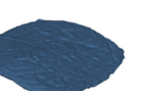
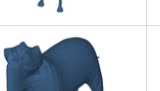
These heterogeneous data do not support an ordinal task-fit ranking. The former 6/6-versus-0/6 table encoded SPPA’s declared capabilities as scoring criteria and is therefore circular; it is excluded from the redesigned evidence. Geometry, latency, complexity, persistence, and input provenance are reported as separate measurements. Figure 3 remains only a qualitative failure audit: neural generators can be visually richer when given suitable visual evidence, while SPPA occupies a different representation point.

Several contemporary image-to-3D candidates are intentionally excluded from the numeric table. Stable Fast 3D installed in a Python 3.10 environment but emitted no benchmark event before the 20 minute timeout in this pass. SPAR3D installation completed, but model loading failed before inference because the required weights were gated. TRELLIS.2, Pixa3D, Direct3D-S2, TripoSG/Tripo P1, Hunyuan3D 2.1, PartCrafter, and Rodin Gen-2.5 were not executed under the frozen four-image local protocol. These are unresolved coverage gaps or reproducibility outcomes, not measured failures of output quality. Therefore the table is valid only as a bounded SPPA runtime substitutability stress test, not as a July 2026 visual image-to-3D SOTA ranking.

The fast-baseline boundary is narrower than the visual-SOTA boundary. For a fair runtime comparison against SPPA, a method must both be plausibly fast for the target operating point and have a successful local run under the same frozen input/resource protocol. Under that rule, TripoSR and Hunyuan3D-2mini Turbo are the reproduced image-conditioned rows in Table 8. Stable Fast 3D and SPAR3D remain the most important missing fast baselines: external sources report sub-second generation, but the local artifact contains only a timeout/no-event outcome for SF3D and a gated-weights outcome for SPAR3D. TRELLIS.2 reports fast low-resolution H100 timings, but it is not yet a reproduced portable-profile baseline here. The remaining contemporary methods are related visual-SOTA context, not fair fast-runtime rows without local timing provenance. The generated audit `benchmarks/results/sota_fast_method_eligibility.md` records this eligibility decision.

Figure 3 is a qualitative method-behaviour audit across six object cases. Its first row is deliberately a text/tag card, not an image crop, not a proxy image, and not real detection ground truth. The image-conditioned baseline meshes shown below were produced from separate local RGBA proxy crops that are omitted from the grid to avoid making the input row look like an SPPA output. The final benchmark should replace this audit with recorded detection crops, masks, bounding boxes, or a public benchmark reference set, and should report explicit quality metrics. An input-provenance manifest and verifier prevent the figure from being re-labeled as a detector or ground-truth benchmark without new evidence. A separate synthetic YOLO reference manifest currently provides three 2D ground-truth bbox labels (**biker**, **cow**, and **tower**), but those legacy samples are bbox-only for this audit and do not provide readable crops for the six-class visual grid. The point of the figure is narrower: neural generators produce meshes with different fidelity/cost

assumptions, while SPPA produces a bounded actor-like primitive proxy from the semantic label.

|                | car                                                                                 | truck                                                                               | tractor                                                                             | biker                                                                                | cow                                                                                   | tree                                                                                  |
|----------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Text/tag input | text:<br>"car"                                                                      | text:<br>"truck"                                                                    | text:<br>"tractor"                                                                  | text:<br>"biker"                                                                     | text:<br>"cow"                                                                        | text:<br>"tree"                                                                       |
| SPPA           |    |    |    |    |    |    |
| TripoSR        |    |    |    |    |    |    |
| Hunyuan3D      |    |    |    |    |    |    |
| Shap-E         |    |    |    |    |    |    |
| Point-E        |  |  |  |  |  |  |

Text/tag row is label-only; RGBA crop inputs are omitted here to avoid proxy-input bias.

Figure 3: Input-aligned visual audit for the six local stress-test cases. The top row is deliberately text/tag only, not a proxy image and not ground truth. SPPA uses the current deterministic semantic-to-proxy generator in its text/tag contract; Shap-E and Point-E use text prompts/tags; TripoSR and Hunyuan3D use separate RGBA crop inputs that are omitted from this grid because proxy crops can visually resemble SPPA outputs. Meshes are rendered in a canonical inspection frame; this changes only the viewer orientation, not the generated geometry. This is a qualitative audit artifact, not a SOTA ranking.

Figure 4 adds four real-image stress probes for the cyclist, electrical-tower, tractor, and tractor-with-trailer cases. These user-supplied crops are real input evidence, not 3D ground truth. The test is deliberately dual-input. In the image path, a YOLOE-26s open-vocabulary detector reads the real image and produces approximate semantic evidence. In the tag-only path, SPPA receives a semantic tag/text input without using the image crop. In both cases the semantic text is routed through a SPPA normalizer before proxy selection, so the paper does not depend on YOLOE producing an exact object name. The broad comparison grid therefore shows a single final SPPA output for each case, while a separate ablation below reports the text-only, detector+metric, and detector+metric+visual variants. Detector masks, bboxes, and declared replay geometry are retained as evidence for audit, scale, pose, and uncertainty, but the full SOTA grid does not show them as a separate competing SPPA method. A separate input-mode ablation below isolates their effect inside the same SPPA generator: text-only selects prior dimensions, detector+metric evidence changes scale/family, and detector+metric+visual evidence adds only budgeted role-conditioned geometry when generic image-space cues support existing roles. When

color evidence is used, it is a conservative chromatic median inside the detector evidence and is applied only to observable semantic material roles. It is not texture recovery, instance segmentation truth, or a visual-quality claim. For example, YOLOE only labels the tractor-with-trailer crop as **vehicle**. Detector-only normalization maps that weak label to a conservative heavy-vehicle proxy; when the same detector evidence is combined with its long metric footprint, SPPA refines it to **articulated\_vehicle**. The reviewed text input **tractor\_trailer** remains the more specific text-conditioned path and is reported separately. This is deliberately not a claim that the image path has identified the exact tractor or trailer type. Shap-E and Point-E receive tag/text prompts; TripoSR and Hunyuan3D receive the same low-quality image crops. The result should be read as an input-quality and semantic-normalization boundary check, not as a visual SOTA ranking.

Figure 4: Real input probes for cyclist, electrical tower, tractor, and tractor-with-trailer cases. The first column is real input evidence, not ground truth. The detector path records YOLOE semantic evidence and reviewed 2D crop provenance. SPPA is shown once as the complete semantic proxy output, while YOLOE mask/ bbox evidence and declared replay geometry are audited separately instead of being displayed as a second SPPA variant. Shap-E and Point-E use tag/text prompts; TripoSR and Hunyuan3D use image crops. This artifact demonstrates why the protocol must separate detector evidence, reviewed semantic text, crop annotations, and ground-truth 3D references.

Table 9: Real-input SPPA observation-fusion audit. Detector confidence (Conf.) is the selected YOLOE detection score. Raw replay dimensions come from YOLOE mask/bbox projection under declared assumed-flight telemetry; SPPA dimensions are the selected constraint-fused proxy dimensions.

| Case                | Conf. | Raw L/W/H (m)        | SPPA L/W/H (m)      | Gate                 | Pose | Tris | Boundary   |
|---------------------|-------|----------------------|---------------------|----------------------|------|------|------------|
| biker               | 0.40  | 3.59 x 1.77 x 1.85   | 1.85 x 0.73 x 1.85  | soft low-conf fusion | no   | 1036 | scale only |
| tower               | 0.46  | 12.77 x 2.89 x 28.00 | 5.60 x 2.89 x 28.00 | height-only fusion   | no   | 270  | scale only |
| tractor             | 0.52  | 4.95 x 4.32 x 2.60   | 4.52 x 2.38 x 2.60  | soft aspect fusion   | no   | 880  | scale only |
| tractor_<br>trailer | 0.47  | 16.85 x 6.21 x 3.40  | 12.11 x 3.76 x 3.40 | soft low-conf fusion | no   | 1904 | scale only |

Table 10: Real-probe semantic normalization. The image-side detector evidence is not treated as exact ground truth; SPPA maps it to a proxy family, optionally refining weak detector labels when metric observation supports a safer conservative shape.

| Probe                      | YOLOE image evidence | Detector-only family             | Detector+observation family               | SPPA reviewed text |
|----------------------------|----------------------|----------------------------------|-------------------------------------------|--------------------|
| Cyclist road image         | person + motorcycle  | two_wheeled_rider → biker        | two_wheeled_rider → biker                 | biker              |
| Mountain tower image       | electric pylon       | power_tower → vertical_structure | power_tower → vertical_structure          | tower              |
| Mountain tractor image     | agricultural vehicle | farm_vehicle → farm_vehicle      | farm_vehicle → farm_vehicle               | tractor            |
| Tractor with trailer image | vehicle              | generic_vehicle → heavy_vehicle  | articulated_vehicle → articulated_vehicle | tractor+trailer    |

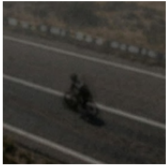
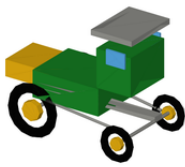
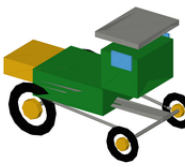
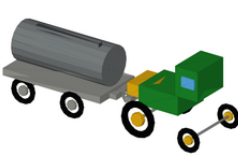
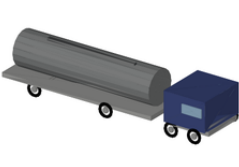
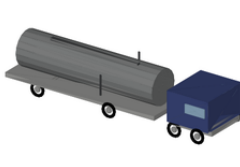
The dual-input benchmark is now frozen as an explicit artifact rather than a visual impression. The image track starts from YOLOE detector evidence on the real images and writes detector crops, confidence, and SPPA-normalized families. The tag/text track receives no image: SPPA, Shap-E, and Point-E use only the reviewed semantic phrase. Table 11 summarizes the bounded systems claim supported by this artifact. It remains separate from any full visual image-to-3D leaderboard because no 3D reference meshes or completed human-preference scores exist for these real images.



Table 11: Frozen real-image and tag/text dual-input benchmark for the bounded SPPA systems claim. The image track uses YOLOE detector evidence and crops for image-to-3D baselines; SPPA additionally reports an observation-refined detector family when geometry supports a safer proxy. The tag/text track uses the reviewed word or phrase for SPPA and text-conditioned baselines. This is not a full visual image-to-3D leaderboard.

| Case            | YOLOE image input                  | SPPA family path                                                                                                                                                               | Image-track methods    | Tag/text methods      |
|-----------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------|
| biker           | <b>person + motorcycle</b> (0.40)  | detector: <b>two_wheeled_rider</b> → <b>biker</b> ; obs: <b>two_wheeled_rider</b> → <b>biker</b> ; text: <b>biker</b> → <b>biker</b>                                           | Hunyuan2-mini, TripoSR | Point-E, Shap-E, SPPA |
| tower           | <b>electric pylon</b> (0.46)       | detector: <b>power_tower</b> → <b>vertical_structure</b> ; obs: <b>power_tower</b> → <b>vertical_structure</b> ; text: <b>tower</b> → <b>tower</b>                             | Hunyuan2-mini, TripoSR | Point-E, Shap-E, SPPA |
| tractor         | <b>agricultural vehicle</b> (0.52) | detector: <b>farm_vehicle</b> → <b>farm_vehicle</b> ; obs: <b>farm_vehicle</b> → <b>farm_vehicle</b> ; text: <b>tractor</b> → <b>tractor</b>                                   | Hunyuan2-mini, TripoSR | Point-E, Shap-E, SPPA |
| tractor+trailer | <b>vehicle</b> (0.47)              | detector: <b>generic_vehicle</b> → <b>heavy_vehicle</b> ; obs: <b>articulated_vehicle</b> → <b>articulated_vehicle</b> ; text: <b>tractor_trailer</b> → <b>tractor_trailer</b> | Hunyuan2-mini, TripoSR | Point-E, Shap-E, SPPA |

Figure 5 and Table 12 isolate the same SPPA generator under three input contracts. The text-only branch receives only the reviewed phrase. The detector+metric branch receives real YOLOE detector text and declared replay scale. The visual branch adds agnostic image-space primitive cues and optional observed material color, but those cues may only condition existing semantic roles; they cannot introduce a new class, texture, or ground-truth part segmentation. This makes the image-side claim stronger and narrower: visual evidence now causes a measured geometry delta in all four cases, while the additional cost remains 20–40 triangles for the cyclist, tower, and tractor cases and 24 triangles for the articulated vehicle case.

|                 | Input crop<br>real image, not GT                                                                                    | Text/tag only<br>reviewed phrase                                                                                                                | YOLOE + metric<br>real detector + replay                                                                                                              | YOLOE + metric + visual<br>adds agnostic cues                                                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| biker           | <br>real crop<br>no 3D reference   | <br>biker<br>1.70x0.60x1.85<br>1,036 tris   5.8 ms             | <br>biker<br>1.85x0.73x1.85<br>1,036 tris   2.4 ms                  | <br>biker<br>1.85x0.73x1.85<br>1,056 tris   3.0 ms<br>visual +20 tris                  |
| tower           | <br>real crop<br>no 3D reference   | <br>tower<br>1.20x1.20x6.00<br>396 tris   1.6 ms               | <br>vertical_structure<br>5.60x2.89x28.00<br>396 tris   3.0 ms      | <br>vertical_structure<br>5.60x2.89x28.00<br>436 tris   1.6 ms<br>visual +40 tris      |
| tractor         | <br>real crop<br>no 3D reference   | <br>tractor<br>3.80x2.00x2.40<br>1,056 tris   2.4 ms           | <br>farm_vehicle<br>4.52x2.38x2.60<br>1,056 tris   2.5 ms           | <br>farm_vehicle<br>4.52x2.38x2.60<br>1,096 tris   2.7 ms<br>visual +40 tris           |
| tractor+trailer | <br>real crop<br>no 3D reference | <br>tractor_trailer<br>7.60x2.15x2.70<br>1,988 tris   3.5 ms | <br>articulated_vehicle<br>13.98x3.54x3.40<br>1,848 tris   3.1 ms | <br>articulated_vehicle<br>13.98x3.54x3.40<br>1,872 tris   3.3 ms<br>visual +24 tris |

Same SPPA generator under three input contracts. Visual cues condition only existing semantic roles; they do not create new classes, ground-truth masks, or dense reconstructions.

Figure 5: SPPA input-mode ablation on the four real-image probes. Each row uses the same deterministic generator under three evidence contracts. Text-only activates a reviewed recipe and prior dimensions. YOLOE+metric uses detector text plus declared replay scale. YOLOE+metric+visual adds only low-cost role-conditioned primitives when agnostic image-space cues support existing roles. This is an ablation of SPPA evidence handling, not ground-truth 3D evaluation.

Table 12: SPPA input-mode comparison. All rows call the same deterministic generator. Text-only uses no image; detector rows use real YOLOE text with declared metric replay; the visual row adds agnostic image-space primitive cues that may condition existing roles.

| Case            | Mode              | Semantic        | Dims L/W/H (m)  | Visual tris | $\Delta$ vs metric | Tris |
|-----------------|-------------------|-----------------|-----------------|-------------|--------------------|------|
| biker           | text              | biker           | 1.70/0.60/1.85  | -           | yes                | 1036 |
| biker           | det+metric        | biker           | 1.85/0.73/1.85  | -           | no                 | 1036 |
| biker           | det+metric+visual | biker           | 1.85/0.73/1.85  | 20          | yes                | 1056 |
| tower           | text              | tower           | 1.20/1.20/6.00  | -           | yes                | 396  |
| tower           | det+metric        | vertical        | 5.60/2.89/28.00 | -           | no                 | 396  |
| tower           | det+metric+visual | vertical        | 5.60/2.89/28.00 | 40          | yes                | 436  |
| tractor         | text              | tractor         | 3.80/2.00/2.40  | -           | yes                | 1056 |
| tractor         | det+metric        | farm            | 4.52/2.38/2.60  | -           | no                 | 1056 |
| tractor         | det+metric+visual | farm            | 4.52/2.38/2.60  | 40          | yes                | 1096 |
| tractor+trailer | text              | tractor+trailer | 7.60/2.15/2.70  | -           | yes                | 1988 |
| tractor+trailer | det+metric        | articulated     | 13.98/3.54/3.40 | -           | no                 | 1848 |
| tractor+trailer | det+metric+visual | articulated     | 13.98/3.54/3.40 | 24          | yes                | 1872 |

To avoid turning Figure 5 into a subjective visual argument, we also score the same rows by bounded evidence-channel coverage. The score counts only audited SPPA channels: semantic normalization, metric replay, visual role cues, yaw gating, and observed material color. It does not score mesh beauty, human preference, or 3D ground-truth accuracy. Under that narrower metric, the reviewed text branch consumes one channel, the detector+metric branch consumes two channels in all four cases, and the detector+metric+visual branch consumes four channels in three cases and five in the tractor case where conservative observed color is also applied. All rows remain inside the lightweight runtime budget. Timing values in Table 13 are wall-clock milliseconds: 3.7 ms corresponds to 0.0037 s, not 3.7 s.

Table 13: SPPA evidence-channel coverage for the real input-mode ablation. The count reports which bounded evidence channels are consumed by the same SPPA generator; it is not a 3D quality score or visual SOTA ranking. Wall time is reported in milliseconds.

| Case            | Mode              | Semantic            | Sem. | Metric | Visual | Yaw | Mat. | Ch. | Budget | Wall (ms) | Tris |
|-----------------|-------------------|---------------------|------|--------|--------|-----|------|-----|--------|-----------|------|
| biker           | text              | biker               | yes  | –      | –      | –   | –    | 1   | pass   | 5.8       | 1036 |
| biker           | det+metric        | biker               | yes  | yes    | –      | –   | –    | 2   | pass   | 2.4       | 1036 |
| biker           | det+metric+visual | biker               | yes  | yes    | yes    | yes | –    | 4   | pass   | 3.0       | 1056 |
| tower           | text              | tower               | yes  | –      | –      | –   | –    | 1   | pass   | 1.6       | 396  |
| tower           | det+metric        | vertical_structure  | yes  | yes    | –      | –   | –    | 2   | pass   | 3.0       | 396  |
| tower           | det+metric+visual | vertical_structure  | yes  | yes    | yes    | yes | –    | 4   | pass   | 1.6       | 436  |
| tractor         | text              | tractor             | yes  | –      | –      | –   | –    | 1   | pass   | 2.4       | 1056 |
| tractor         | det+metric        | farm_vehicle        | yes  | yes    | –      | –   | –    | 2   | pass   | 2.5       | 1056 |
| tractor         | det+metric+visual | farm_vehicle        | yes  | yes    | yes    | yes | yes  | 5   | pass   | 2.7       | 1096 |
| tractor_trailer | text              | tractor_trailer     | yes  | –      | –      | –   | –    | 1   | pass   | 3.5       | 1988 |
| tractor_trailer | det+metric        | articulated_vehicle | yes  | yes    | –      | –   | –    | 2   | pass   | 3.1       | 1848 |
| tractor_trailer | det+metric+visual | articulated_vehicle | yes  | yes    | yes    | yes | –    | 4   | pass   | 3.3       | 1872 |

Table 14 isolates the more ambitious image-side SPPA route. This route does not use the reviewed tag; it starts from YOLOE detector text and metric replay evidence. The important boundary is visible in the tractor-with-trailer case: the system refines **vehicle** to a conservative **articulated\_vehicle**, not to the more specific reviewed **tractor\_trailer** class.

Table 14: Detector-plus-observation SPPA route on frozen real probes. This route uses YOLOE semantic evidence plus replay metric dimensions, without the reviewed text tag, and remains bounded by the lightweight proxy budget.

| Case            | Detector+obs SPPA                                       | Rule                             | ms    | Tris | Boundary            |
|-----------------|---------------------------------------------------------|----------------------------------|-------|------|---------------------|
| biker           | <b>biker</b> → <b>biker</b>                             | person+two-wheel                 | 3.528 | 1036 | conservative family |
| tower           | <b>vertical_structure</b> → <b>vertical_structure</b>   | power infrastructure             | 1.372 | 270  | conservative family |
| tractor         | <b>farm_vehicle</b> → <b>farm_vehicle</b>               | farm vehicle                     | 2.477 | 936  | conservative family |
| tractor+trailer | <b>articulated_vehicle</b> → <b>articulated_vehicle</b> | long-footprint metric refinement | 2.963 | 1848 | conservative family |

We also run a metric proxy replay on the same four real images (Table 19). This replay uses the real YOLOE detector outputs above, but injects declared UAV pose, AGL, camera mount, and family height priors as scenario parameters. It is therefore a real-image and real-detector test of the SPPA observation-to-descriptor path, not a measured flight-localization result. The replay now emits a **SPPA-OBS-0.1** identifier and descriptor uncertainty metadata for each case; those fields document the scenario-relative status rather than upgrading the replay to ground truth. The detector artifact now stores native YOLOE/Ultralytics mask polygons for the selected detections, so the replay can exercise the native detector-mask path rather than only a bbox or CV-derived silhouette proxy. These masks are still detector evidence, not ground-truth segmentations. The full-frame detector evidence for these four cases is included in the zoomed audit in Supplementary Figure S1.

To test the user’s intended bridge from detector evidence to proxy parts without injecting class-specific part layouts, we added an experimental agnostic image-space primitive probe (Figure 6). The fitter receives only real crop pixels, the detector bbox, and unlabeled mask polygons; detector labels and reviewed SPPA tags are retained only for audit. It extracts generic cues such as round primitive pairs, coherent line structure, principal axes, and mask envelopes. The intended unit is therefore any detector crop, not a cyclist-specific or vehicle-specific rule: the same procedure is applied to all objects and can only report generic visual primitive evidence that is actually visible in the crop. On the four real images, the probe finds a strong round-pair cue for the cyclist and tractor, a coherent multi-line cue for the tower, and a multi-line cue for the tractor/trailer case while retaining raw circular primitives only as audit evidence. This supports a more ambitious image-to-primitive research path, but it is not yet a claim of universal part recovery or ground-truth 3D reconstruction. The current SPPA descriptor now consumes this evidence in a role-conditioned way (Table 15): generic round and line cues can annotate existing roles such as tires, frame elements, tower metal, attachments, or container details. The descriptor also stores a compact image-space geometry profile—for example round-pair axis/radius ratios or dominant line orientation—once per object, so the information is available for downstream pose/scale reasoning without duplicating it in every primitive. When no metric yaw, telemetry heading, track velocity, or mask-PCA yaw is available, SPPA also records an axial image-space orientation from the same profile. This orientation is explicitly marked as image-space and ambiguous; it is not a world yaw until a camera/ground projection module consumes it. These cues cannot create a new semantic class or add mesh topology. This is a deliberately bounded bridge from pixels to proxy structure.

Table 15: SPPA visual part evidence audit. Generic pixel cues from the detector crop annotate existing descriptor roles, store a compact image-space geometry profile, and may add budgeted conditioning primitives to existing roles. They do not change the semantic class or claim ground-truth part segmentation.

| Case            | Generic cue scope    | Supported descriptor roles   | Visual tris | ms    | Tris | Boundary            |
|-----------------|----------------------|------------------------------|-------------|-------|------|---------------------|
| biker           | round-pair candidate | frame, hub, tire             | 20          | 3.573 | 1056 | existing roles only |
| tower           | multi-line candidate | tower metal                  | 40          | 1.270 | 436  | existing roles only |
| tractor         | round-pair candidate | attachment, hub, tire        | 40          | 2.190 | 1096 | existing roles only |
| tractor+trailer | multi-line candidate | container detail, attachment | 24          | 3.296 | 2012 | existing roles only |

The visual orientation field is also checked against the UAV metric projection branch (Table 16). This comparison is deliberately not used as a hidden correction step: raw pixel yaw and projected ground-footprint yaw are independent ambiguous axial cues with different error sources. The fair comparison therefore projects the visual axis endpoints (wheel-pair centers or the longest image-space line) through the same declared UAV camera model before comparing them with the footprint yaw. With this projection-aware gate the four real probes produce no strong divergence: two axes are aligned and two are weakly aligned under the declared replay geometry. SPPA still keeps the metric yaw tied to camera/flight projection and uses the projected visual axis only as support/gating evidence. When the gate is non-divergent, the descriptor selects the axial projected-footprint yaw in the declared replay frame; it does not treat the raw image-space axis as world yaw and does not claim ground truth. This makes the current system more conservative, but also more defensible: it shows how visible primitives can support pose reasoning only after the UAV projection model is in the loop. The descriptor schema and submission precheck enforce this as a conditional contract: whenever `visual_metric_yaw_consistency` is declared as evidence, the projected visual-axis gate, axial footprint-yaw selection, coordinate frame, ambiguity flag, and policy identifier must be present in the descriptor.

Table 16: SPPA visual–metric yaw gate on real image probes. Pixel-space visual axes are projected through the declared UAV replay camera model before comparison with the projected ground footprint. When the gate is non-divergent, SPPA selects the axial projected-footprint yaw; both projected quantities remain scenario-relative, not measured telemetry or ground truth.

| Case            | Visual axis | Footprint yaw | $\Delta$ | Agreement      | Use in SPPA                  |
|-----------------|-------------|---------------|----------|----------------|------------------------------|
| biker           | 4.8°        | 1.6°          | 3.2°     | aligned        | selected axial footprint yaw |
| tower           | 13.3°       | 8.6°          | 4.6°     | aligned        | selected axial footprint yaw |
| tractor         | 108.3°      | 80.1°         | 28.3°    | weakly aligned | selected axial footprint yaw |
| tractor+trailer | 49.3°       | 77.4°         | 28.1°    | weakly aligned | selected axial footprint yaw |

Supplementary Figure S2 shows the same bridge visually: the crop-level evidence is generic and unlabeled, the descriptor records only role support plus a compact image-space profile and, when available, a projection-gated axial footprint yaw; the proxy render remains the lightweight SPPA mesh. This is the current honest ambition boundary: SPPA uses visual evidence to condition parts and gate orientation fields that already belong to the normalized proxy, without claiming dense single-image reconstruction, detector-supervised part segmentation, or ground-truth world-frame yaw.

A connector regression guards the main failure mode visible in naive primitive generators: disconnected pieces. Table 17 reports the generated descriptor counts and length ranges. This remains a lightweight semantic-connectivity claim, not a photorealistic reconstruction claim.

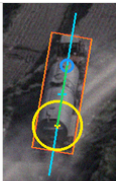
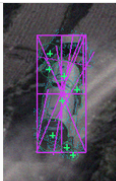
Table 17: SPPA connector-constraint regression. Explicit lightweight connectors are required for semantic parts that should touch; this is a connectivity contract, not a photorealistic reconstruction claim.

| Case            | Connectors | Roles                                  | Length range   | Status |
|-----------------|------------|----------------------------------------|----------------|--------|
| biker           | 16         | bike frame, rider clothing, rider skin | 0.155–1.332 m  | passed |
| tower           | 19         | vertical structure metal               | 2.773–24.680 m | passed |
| tractor         | 6          | vehicle metal or hub                   | 2.151–3.436 m  | passed |
| tractor+trailer | 6          | vehicle metal or hub                   | 0.819–3.502 m  | passed |

The anti-shortcut and robustness battery is summarized in Table 18. The checks mutate labels, case identities, image paths, detector-mask ordering and duplicates, combined non-geometric side channels, mirror transforms, and deterministic photometric variants. Synthetic controls, a deterministic sweep, and a four-seed fuzz test exercise known primitive cases. Primary decisions pass in every row; mirror and photometric checks still retain secondary audit warnings for line-support and weak-pair drift. This is the intended boundary: the battery constrains shortcuts and regression failures in the visual bridge, but it does not prove universal primitive correctness or real UAV illumination rates.

Table 18: Agnostic visual-bridge robustness battery. These checks support anti-shortcut and regression claims for the frozen real probes and synthetic controls; they do not claim universal real-world primitive correctness.

| Check                   | Scope      | N   | Metric                                        | Status | Warn. | Boundary                                                          |
|-------------------------|------------|-----|-----------------------------------------------|--------|-------|-------------------------------------------------------------------|
| Cue extraction          | real crops | 4   | 4 frozen real probes                          | passed | 0     | Checks descriptor/export consistency, not primitive correctness.  |
| Label invariance        | real crops | 4   | geometry hash unchanged                       | passed | 0     | Guards label/tag shortcuts only.                                  |
| Identity invariance     | real crops | 4   | geometry hash unchanged                       | passed | 0     | Guards case-name hardcoding only.                                 |
| Path invariance         | real crops | 4   | geometry hash unchanged                       | passed | 0     | Guards filename/path shortcuts only.                              |
| Detector representation | real crops | 12  | mask order/duplicates unchanged               | passed | 0     | Guards representation-only side channels.                         |
| Combined side channels  | real crops | 4   | geometry hash unchanged                       | passed | 0     | Does not prove primitive correctness.                             |
| Mirror equivariance     | real crops | 4   | primary stable, 17 secondary warnings         | passed | 17    | Secondary line/weak-pair drift remains recorded.                  |
| Photometric stability   | real crops | 16  | primary stable, 15 secondary warnings         | passed | 15    | Deterministic perturbations only, not real illumination coverage. |
| Synthetic controls      | synthetic  | 5   | all controls passed                           | passed | 0     | Known geometry controls, not detector quality.                    |
| Synthetic sweep         | synthetic  | 72  | acc=1.000, round F1=1.000, line F1=1.000      | passed | 0     | Synthetic primitive behavior only.                                |
| Synthetic fuzz          | synthetic  | 240 | 4 seeds, acc=1.000, round/line F1=1.000/1.000 | passed | 0     | Randomized synthetic cues, not real UAV rates.                    |

| <b>Crop</b><br>real pixels                                                         | <b>Mask</b><br>detector evidence                                                   | <b>Mask fit</b><br>bbox/mask only                                                  | <b>Image cues</b><br>pixels+bbbox+mask, no label                                    | <b>Audit</b><br>anti-bias metrics                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>biker</b><br>source crop                                                        | <b>biker</b><br>unlabeled masks                                                    | <b>biker</b><br>geometry-only mask fit                                             | <b>biker</b><br>generic primitive cues                                              | <b>biker</b><br>no semantic filter input                                                                                                                                                                                                                     |
|   |   |   |   | <b>biker</b><br>label used: false<br>mask scope: unlabeled_component_candic<br>image scope: round_part_pair_candidate<br>round raw: 8 pairs: 3<br>lines: 16 coherent: false<br>edge density: 0.11424<br>grade: experimental_image_part_candidat              |
| <b>tower</b><br>source crop                                                        | <b>tower</b><br>unlabeled masks                                                    | <b>tower</b><br>geometry-only mask fit                                             | <b>tower</b><br>generic primitive cues                                              | <b>tower</b><br>no semantic filter input                                                                                                                                                                                                                     |
|   |   |   |   | <b>tower</b><br>label used: false<br>mask scope: axis_aligned_envelope_only<br>image scope: multi_line_structure_candida<br>round raw: 8 pairs: 0<br>lines: 16 coherent: true<br>edge density: 0.23511<br>grade: experimental_image_part_candidat            |
| <b>tractor</b><br>source crop                                                      | <b>tractor</b><br>unlabeled masks                                                  | <b>tractor</b><br>geometry-only mask fit                                           | <b>tractor</b><br>generic primitive cues                                            | <b>tractor</b><br>no semantic filter input                                                                                                                                                                                                                   |
|   |   |   |  | <b>tractor</b><br>label used: false<br>mask scope: single_blob_envelope_only<br>image scope: round_part_pair_candidate<br>round raw: 8 pairs: 3<br>lines: 16 coherent: false<br>edge density: 0.28921<br>grade: experimental_image_part_candidat             |
| <b>tractor_trailer</b><br>source crop                                              | <b>tractor_trailer</b><br>unlabeled masks                                          | <b>tractor_trailer</b><br>geometry-only mask fit                                   | <b>tractor_trailer</b><br>generic primitive cues                                    | <b>tractor_trailer</b><br>no semantic filter input                                                                                                                                                                                                           |
|  |  |  |  | <b>tractor_trailer</b><br>label used: false<br>mask scope: axis_aligned_envelope_only<br>image scope: multi_line_structure_candida<br>round raw: 8 pairs: 0<br>lines: 16 coherent: false<br>edge density: 0.21675<br>grade: experimental_image_part_candidat |

Agnostic probe: labels/tags are audit-only; primitive cues are extracted from pixels, detector bbox, and unlabeled masks for every object.

Figure 6: Experimental agnostic image-space primitive probe. The method uses real pixels, detector bboxes, and unlabeled masks for every object, while semantic labels are audit-only. Green circles indicate validated round-pair candidates when the geometry supports them; magenta lines show generic line evidence from edges plus bbox/mask/PCA support lines used for deterministic stability. The right column records the anti-label audit and confidence grade for each case. This is an analysis probe for future image-conditioned part cues, not the runtime generator used for the SPPA outputs in Figure 4.

Table 19: Real-image SPPA metric replay using bbox plus native YOLOE mask polygon. Images and YOLOE detections are real local inputs; flight telemetry and height priors are declared replay parameters, so position and scale are scenario-relative rather than measured ground truth.

| Case            | YOLOE evidence              | SPPA proxy                | Scenario dims (m)                | Local pose (m)          |
|-----------------|-----------------------------|---------------------------|----------------------------------|-------------------------|
| Cyclist         | <b>person + motorcycle</b>  | <b>biker</b>              | $3.59 \times 1.77 \times 1.85$   | $N = -3.67, E = -1.75$  |
| Tower           | <b>electric pylon</b>       | <b>vertical_structure</b> | $12.77 \times 2.89 \times 28.00$ | $N = -1.72, E = 0.56$   |
| Tractor         | <b>agricultural vehicle</b> | <b>farm_vehicle</b>       | $4.95 \times 4.32 \times 2.60$   | $N = -8.45, E = -10.96$ |
| Tractor/trailer | <b>vehicle</b>              | <b>heavy_vehicle</b>      | $16.85 \times 6.21 \times 3.40$  | $N = -2.38, E = -7.61$  |

A dedicated replay verifier checks this native-mask path. For the four real images, the selected YOLOE detections all include native mask polygons with 12, 35, 33, and 54 points respectively. The replay bridge projects those masks into candidate oriented footprints before the SPPA constraint-fusion gate decides which scale, footprint, and pose components are safe to carry into the selected runtime proxy. This verifies the detector-evidence data path and byte-budget handling; it still does not verify mask correctness against ground-truth segmentation, and the final descriptor may omit unsafe mask-derived pose.

The controlled SPPA contract benchmark isolates the difference between a label-only semantic prior baseline and the current metric-observation path. Synthetic masks are generated from declared object dimensions under a nadir pinhole camera, then compiled through the SPPA observation and descriptor contracts. This is a regression of contract behavior, not a real UAV segmentation benchmark. Under this controlled setting, the label-only baseline had summed dimension-error P50/P95 of 2.400/13.900 m; the current metric-observation path had 0.000/0.000 m. Mean improvement was 4.653 m, with descriptor build P50/P95 of 372.0/460.5  $\mu$ s and a maximum of 1,692 triangles.

Table 20: SPPA controlled contract benchmark. The baseline is the semantic prior without metric observation; the current path uses synthetic metric mask evidence through **SPPA-OBS-0.1**. This supports scale-adaptation plumbing only, not real detector-mask quality or real flight calibration.

| Case               | Label                       | Baseline err. (m) | Current err. (m) | Improve (m) | Source                   |
|--------------------|-----------------------------|-------------------|------------------|-------------|--------------------------|
| Cyclist long view  | <b>cyclist</b>              | 0.470             | 0.000            | 0.470       | mask-projected footprint |
| Tower tall proxy   | <b>electric pylon</b>       | 13.900            | 0.000            | 13.900      | mask-projected footprint |
| Tractor wide       | <b>agricultural vehicle</b> | 7.850             | 0.000            | 7.850       | mask-projected footprint |
| Truck/trailer long | <b>truck</b>                | 2.400             | 0.000            | 2.400       | mask-projected footprint |
| Cow close          | <b>cow</b>                  | 2.450             | 0.000            | 2.450       | mask-projected footprint |
| Car small          | <b>car</b>                  | 0.850             | 0.000            | 0.850       | mask-projected footprint |

A second SPPA benchmark isolates the value of mask-footprint evidence relative to axis-aligned bboxes. We generate 162 synthetic rotated rectangles across three labels, three sizes, three aspect ratios, and six axial rotations under a known nadir camera, then run both paths through the same



SPPA-OBS-0.1 bridge. The bbox path over-estimates rotated footprints: summed dimensional error is 2.417/6.407 m at P50/P95 and axial-yaw error is 15.0/45.0°. The mask/silhouette path has 0.000/0.000 m error against the supplied synthetic polygon reference and preserves the polygon in 100% of generated descriptors. Non-axis-aligned median improvement is 2.951 m. This supports the contract claim that silhouette evidence prevents bbox inflation when such evidence exists; it is still not a claim that the real YOLOE replay above has ground-truth masks.

Table 21: SPPA bbox-vs-silhouette projection benchmark under known synthetic camera geometry. The mask path uses supplied synthetic polygons; real detector mask quality is outside this table.

| Path                       | Cases | Dim. err. P50/P95 (m) | Yaw err. P50/P95 | Descriptor evidence                        |
|----------------------------|-------|-----------------------|------------------|--------------------------------------------|
| Bbox-only projection       | 162   | 2.417/6.407           | 15.0/45.0°       | axis-aligned bbox                          |
| Mask/silhouette projection | 162   | 0.000/0.000           | 0.0/0.0°         | polygon preserved in descriptor            |
| Non-axis improvement       | 135   | 2.951/6.407           | –                | bbox inflation removed when polygon exists |

These results narrow the argument. TripoSR and Hunyuan3D are better choices when the goal is high-fidelity asset generation from a usable image crop. Shap-E and Point-E demonstrate that text-conditioned meshes can be produced from the same tag path, but their outputs are not track-persistent proxy contracts. SPPA’s measured advantage is different: no GPU inference, millisecond-scale local proxy creation, bounded primitive geometry, and per-track updateability. Whether that advantage matters to pilots is still unproven.

## 10.5 Role-Preservation Diagnostic

The main competitor set for SPPA is the set of operational representation alternatives in Table 1, especially curated assets and fast text/image-to-3D generators. Geometry-only displays are retained only as negative controls and lower-bound runtime references; construction timings for isolated boxes, capsules, or ellipsoids are therefore not reported as a result.

The implemented diagnostic asks whether SPPA preserves semantic part roles while adapting dimensions. For four synthetic vehicle variants, median dimension error was 0.3926 for fixed templates, 0.0000 for trivial global scaling, and 0.0223 for SPPA parametric vehicles. The global-scaling result is deliberately not a win condition: it fits the outer box by deforming all parts together, including cab and wheels. The SPPA diagnostic is therefore not a morphology benchmark. Its supported claim is narrower: role-assigned parts such as cargo, cab, windows, and wheels can follow different update rules instead of collapsing into a scaled copy of the same object. A dedicated short/long truck invariance check makes this concrete: increasing vehicle length from 5.2 m to 8.2 m kept cab scale delta at 0.0, tire scale delta at 0.0, increased cargo length by 2.925 m, and increased tire count from 6 to 8.

A calibrated-mask variant now exercises the same path from observed footprint evidence rather than manually supplied dimensions. Over 64 synthetic rectangle masks with explicit ground scale, the calibrated mask-to-shape benchmark derived metric length/width with 0.000/0.000 m P50/P95 summed dimensional error and kept the uncalibrated control path at `template_prior`. Local geometry build time was 442/480  $\mu$ s at P50/P95, and descriptor build time was 297/321  $\mu$ s at P50/P95. This verifies only the deterministic contract path *mask polygon + supplied ground scale*  $\rightarrow$  *parametric parts*; it is not evidence of real UAV segmentation quality, camera calibration accuracy, or silhouette-to-part assignment under occlusion.

## 10.6 Descriptor and Unreal Runtime Tests

The descriptor/update benchmark generated descriptors for exact labels, keyword-resolved labels, and unsupported labels. Exact and keyword labels resolved to known finite archetypes; unsupported labels produced unknown fallback descriptors. In the replay benchmark, descriptor build

with parts over 81,568 threshold-expanded rows had P50/P95 of 391.8/515.0  $\mu$ s. The P99 was 671.2  $\mu$ s, but the maximum descriptor-build observation was 360.050 ms. Scheduler decisions had P50/P95/P99 of 9.2/14.2/19.2  $\mu$ s, and update-packet builds had P50/P95/P99 of 16.8/75.3/91.0  $\mu$ s. Contract validation passed 453 descriptor and update-packet checks. These are Python contract measurements, not Unreal frame time; they support typical-case local contract cost, not a hard real-time guarantee.

A preliminary link-budget model over the same replay separates byte accounting from radio-link validation. If no-op packets are suppressed, measured SPPA JSON update traffic was 25.8–37.4 kB/s across scheduler thresholds; emitting every packet was 78.5–94.0 kB/s. A favorable modeled binary box stream was about 3.27 kB/s, while modeled compressed video at 2/5/10 Mbps corresponds to 250/625/1250 kB/s. This does not show that SPPA is the lowest-bandwidth semantic representation. It shows that SPPA is a richer actor representation with byte cost between sparse boxes and continuous video/mesh transfer under this replay model.

An additional adversarial resolver audit covered 12 compound or misleading labels. Visual-artifact contexts such as **toy truck**, **vehicle shadow**, **dog statue**, and **person poster** were forced to the unknown fallback; equipment-only labels such as **worker helmet** also fell back instead of spawning a person. Compound family labels such as **animal feed trailer**, **horse trailer**, **emergency vehicle**, **road sign**, and **field worker** resolved to reviewed family archetypes. This is a resolver-contract check, not a human-factors validation of fallback interpretation.

The offline recipe-review gate complements this resolver audit. It evaluates candidate language/ontology recipes against the finite primitive vocabulary, known material roles, runtime resolver result, triangle budget, and the rule that no language model may be called in runtime. In the current artifact, the gate accepted a **dump truck** candidate as a reviewed heavy-vehicle mapping, allowed an **unlisted obstacle** only as fallback, and rejected **person poster**, **delivery drone**, and **construction crane** candidates for false semantics, unreviewed topology/material roles, or runtime-LLM dependence. This supports the narrower claim that SPPA has a cache-admission guard for candidate recipes.

A mask-footprint benchmark then evaluated descriptor-level shape evidence over 162 synthetic rectangle masks: three lengths, three aspect ratios, six axial rotations, and three corner-jitter levels. Across clean non-axis-aligned masks, the AABB dimensional error was 50.568/109.808 px at P50/P95, while the PCA oriented footprint recovered 0.000/0.000 px dimensional error and 0.000/0.000 deg axial-yaw error. With jittered non-axis-aligned masks, AABB error was 51.957/114.235 px and oriented-footprint error was 6.615/12.389 px, with 0.158/0.666 deg yaw error. Axis-aligned masks can tie the AABB baseline, and jitter can make AABB slightly better for those aligned cases; the supported claim is therefore limited to extracting oriented image-space footprint evidence from supplied polygon masks, not real UAV silhouette recovery.

Unreal tests then exercised the native backend in three steps. First, an actor-level regression test loaded a real SPPA-DESC-0.2 fixture with 12 parts, generated 12 static-mesh components, propagated material/evidence/uncertainty tags, accepted valid pose updates, rejected mismatched descriptor updates, rejected invalid JSON without replacing the current proxy, and applied a matching-topology **shape\_param\_update** while preserving the same component names. Second, component replay used the same **obstacles** array payload shape as **/api/ui/data**. Third, a local-loopback HTTP replay exercised Unreal’s HTTP polling branch with the same payload. A recorded-payload adapter then converted seven logged **obstacle\_ingest** observations into common asset-compatible, SPPA emit-all, and SPPA changed-only payload streams. Unreal Editor-Cmd consumed those recorded streams through the same component entry point with asset, actor-proxy, and instanced-proxy backends. A later packaged executable run consumed the same streams with rendered frames. This is stronger than a payload-only adapter check, but it remains an internal short-stream replay; it is not live network telemetry, VR, phase-aligned GPU-profiler evidence, or operator validation.

The packaged evidence consists of synthetic density replay, HISM batching diagnostics, and a short recorded-payload replay. The baseline packaged replay, 20260702T122743Z, used three

Table 22: Selected Unreal evidence that constrains the SPPA claim. Values are preliminary and do not include VR headset, live-network, broad curated-asset validation, or phase-aligned GPU-profiler validation. Auxiliary regression checks are kept in the artifact package, not in this summary table.

| Test                                           | Result                                                                                                                          | Main boundary                                                                     | Claim impact                                                                          |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Backend switch smoke                           | Same obstacle input selects assets, actor proxies, or HISM proxies; default remains assets                                      | Editor smoke, not operator workflow                                               | SPPA is optional, not a forced replacement                                            |
| Editor-Cmd recorded replay                     | 27/27 payload applications accepted across asset, actor-proxy, and HISM backends; SPPA emit-all P95 apply: 1.614/3.805/4.049 ms | component ingestion under Editor-Cmd/Null-RHI                                     | recorded streams reach all backend routes                                             |
| Packaged recorded replay 20260703T024800Z      | 27/27 payload applications accepted; frame P95 assets/proxy/HISM: 16.551/6.531/4.809 ms; apply P95: 2.566/5.670/3.246 ms        | one short replay, no live network, no phase-aligned GPU profiler, no headset      | recorded streams run in packaged executable                                           |
| Packaged synthetic 100-object replay           | actor-proxy pose/shape frame P95: 17.814/19.256 ms                                                                              | synthetic desktop obstacles; misses 60 FPS P95                                    | desktop feasibility only, not VR readiness                                            |
| Packaged HISM scaling 20260703T033655Z/033840Z | 11 active draw groups; 100-object no-CSV rerun shape frame P95: 37.126 ms; 250/500 shape P95: 94.018/186.113 ms                 | synthetic desktop obstacles; shape updates miss 30 FPS P95 at 100+ objects        | batching helps draw pressure, but dense shape updates remain limiting                 |
| Packaged HISM partial updates 20260703T040602Z | 10% pose and 10% shape changed-track schedule; 500-object pose/shape frame P95: 16.528/29.787 ms with 11 draw groups            | synthetic scheduler rate, not flight-derived; dense all-object updates still fail | partial-update contract supports changed-only scheduling, not dense all-track updates |
| Packaged cow/tower asset subset                | project assets vs. SPPA shape frame P95: 6.789 vs. 22.445 ms                                                                    | narrow two-label subset                                                           | curated assets remain better when available                                           |

repetitions, 30 warmup frames, and 120 measured frames per phase. At 100 synthetic obstacles, `semantic_proxy` used 432 components, about 208,864 estimated triangles, and 432 estimated draw calls; pose and shape frame P95 were 17.814 ms and 19.256 ms. This passed a 33.3 ms P95 desktop frame budget in that run but missed a 16.7 ms P95 budget, and phase-aligned GPU-profiler counters were not captured. The allowed claim is therefore limited to packaged desktop feasibility in specific measured cases, not VR readiness.

The recorded-payload packaged run, 20260703T024800Z, replayed three externally generated JSONL streams through the same asset/proxy backend switch: common asset-compatible payloads, SPPA emit-all payloads, and SPPA changed-only payloads. All 27 applications were accepted. The HISM route had median active draw-group count 6 and frame P95 4.809 ms in this short replay, while the actor proxy had median component count 31 and frame P95 6.531 ms. This is useful integration evidence, but it is still a short internal replay without live network timing, phase-aligned GPU-profiler counters, headset execution, or operator evidence.

The HISM measurements are a separate opt-in diagnostic family, not a replacement for the baseline replay. In 20260702T215340Z, a one-repetition run with shorter measurement windows, the HISM route reduced the 100-object component and estimated draw-call count from 432 to 11, and reduced `shape_param_update` action P95 from 20.736 ms to 13.270 ms. The 100-object

shape-stream frame P95 improved from 42.903 ms to 33.626 ms, which was still above the 30 FPS P95 budget. The HISM update path was then changed to defer render-state dirty marking during batch instance updates and flush once per touched group. A longer HISM-only packaged run, 20260703T031246Z, used three repetitions, 30 warmup frames, and 120 measured frames per phase at 100 objects. It reported 11 active draw groups, pose-stream frame P95 13.884 ms, shape-stream frame P95 29.715 ms, and shape-stream frame P99 35.021 ms, with 6 of 360 shape frames above 33 ms. This was promising but not robust. A later dense sweep without Unreal CSV profiling, 20260703T033655Z, tested 100, 250, and 500 objects with the same HISM backend. The 100-object shape-stream frame P95 was 38.657 ms, and an isolated 100-object no-CSV rerun, 20260703T033840Z, reported 37.126 ms. At 250 and 500 objects, shape-stream P95 rose to 94.018 ms and 186.113 ms, respectively. The supported conclusion is therefore narrower: HISM batching keeps the proxy scene at 11 active draw groups across these synthetic densities and create-steady frames remain cheap, but stable 30 FPS P95 shape-update performance is not yet demonstrated even at 100 objects. This still does not support a 60 FPS, headset, live-network, or phase-aligned GPU-profiler claim.

The HISM path was then extended with an explicit `partial_obstacle_update` contract: unmentioned entities are retained, while mentioned entities replace their old instance keys before new parts are upserted. The backend verifier creates two entities, applies a partial shape-update to only one, and verifies 6/6 expected live/HISM instances after the untouched entity is preserved. A new packaged run, 20260703T040154Z, confirmed that dense 100-object shape updates still fail (shape-stream P95 46.151 ms). With shape updates limited to 10% of objects but dense pose updates retained, shape-stream P95 stayed below 33.3 ms through 500 objects, but pose-stream P95 reached 52.474/104.502 ms at 250/500 objects. Under a changed-track schedule with both pose and shape updates limited to 10%, 20260703T040602Z reported 500-object pose/shape frame P95 of 16.528/29.787 ms and no measured hitches above 33 ms. The allowed claim is therefore conditional: partial updates support a changed-only scheduler stress case, but the 10% rate is not yet derived from flight logs and does not prove dense all-object update performance.

Current HISM regressions expose `SEMANTIC_PROXY_INSTANCED` as a live opt-in backend, preserve local pose through the component route, reject mismatched descriptor/update packets, support group-level material and uncertainty tags, provide an opt-in collision policy, and support eight per-instance custom-data floats for resolved color, confidence, evidence source, and uncertainty. In an observed-color stress run at 100 objects, per-instance custom data reduced active HISM components/draw calls from 188 to 11 and reduced create/pose/shape action P95 from 27.792/9.669/18.303 ms to 18.608/7.050/15.351 ms, but did not improve the corresponding shape-stream frame P95 (37.725 vs. 38.136 ms). This supports HISM as an engineering direction for batching, not as completed dense-VR validation; phase-aligned GPU-profiler measurements and operational collision semantics remain open. The same backend verifier now includes a local georeferenced-pose regression: direct HISM payloads using `world_m` and `world_cm`, and component-routed payloads using both `SemanticProxy` and `SemanticProxyInstanced`, place a zero-centered descriptor at [1250,-400,120] cm with yaw 37 deg. This is only local-frame pose preservation, not evidence of live UAV georeferencing, camera calibration, Cesium accuracy, or VR runtime behavior.

A second packaged run adds a narrower real-asset baseline for the two labels with reviewed project static meshes in the repository, `cow` and `tower`. At 50 objects, `project_assets` used 50 static-mesh components and about 167,300 estimated triangles; `semantic_proxy` used 200 components and about 113,200 estimated triangles. The real-asset subset had lower P95 frame times for create, pose, and shape phases, with shape-stream P95 6.789 ms versus 22.445 ms for SPPA. This is not a failure of the SPPA claim: it shows that curated assets are preferable when an appropriate asset exists and loads cheaply. SPPA's intended gap is uncovered or mismatched semantic telemetry, not replacement of available high-quality assets.

## 11 Discussion

The confirmatory result answers the preregistered question: under identical calibrated silhouettes and optimizer budget, a frozen family part graph improves lightweight occupancy over a generic eight-part graph by a large and stable margin. The knowledge representation, not a clever optimizer, is the active ingredient. Secondary gains over text-only SPPA show that silhouette evidence helps when the same family graph is held fixed. Gains over boxes, capsules, and billboards are expected but still quantified under the shared protocol.

Several limits remain binding. The held-out set is synthetic and developer-held-out; positive OOD-stratum differences do not license claims about real UAV masks, detectors, or metric flight geometry. Family tokens are supplied, so the experiment is family-conditioned rather than open-set. Nonsemantic visual hull [22] can approach SPPA-MVFit IoU because it is free to materialize denser occupancy without an eight-part role-labeled actor contract; runtime persistence, updateability, and semantic roles are separate properties. Neural image-to-3D generators remain the wrong competitor for sparse telemetry without crops, but they remain preferable for offline asset authoring when budget allows; the measured wave of Section 10.1 quantifies this boundary on the sealed cases rather than by analogy. Unreal/HISM packaged measurements constrain deployment cost but do not establish dense VR readiness, headset performance, or operator benefit. A clean proxy can still look more certain than the underlying evidence; visual uncertainty encoding is specified, not human-validated.

Three upgrade paths are deliberately left to future work because any of them would have invalidated the sealed confirmatory comparison had they been introduced mid-course. First, the frozen coordinate-descent fitter with its preregistered evaluation budget could be replaced by population-based (CMA-ES) or differentiable-rendering gradient fitting, which we expect to add small IoU gains at similar cost; the present paper isolates the representation effect, so the optimizer was held fixed on purpose. Second, the primitive vocabulary could be extended from boxes, ellipsoids, and cylinders to superquadrics; the implicit-OOD stratum already shows the current vocabulary’s ceiling against such sources. Third, the iterative fit could be amortized by a small learned mask-to- $\theta$  predictor, keeping the primitive representation, role labels, and update contract while moving per-case cost from milliseconds toward microseconds. None of these changes alters the claim evaluated here: family conditioning is the active ingredient, and the sealed protocol measures it independently of fitter sophistication.

## 12 Threats to Validity

First, confirmatory geometry is synthetic. Even the implicit-OOD stratum is internally generated, so external public UAV validity is untested. Second, family tokens are observed inputs; the experiment does not evaluate open-set family discovery. Third, silhouettes are clean orthographic projections plus prespecified corruptions, not real detector masks. Fourth, secondary systems evidence (Unreal, YOLOE real probes, neural-generator grids) uses different contracts and must not be pooled into a single quality score. Fifth, timing is local CPU wall-clock for the fitter, not flight-laptop Unreal frame time. Sixth, the Holms-style secondary  $p$ -values in the analysis artifact are not used in this manuscript; secondary inference is reported via bootstrap intervals only.

## 13 Claim Boundaries

Supported:

- On the sealed developer-held-out synthetic test, family-conditioned SPPA-MVFit improves clean voxel IoU over equal-budget Generic-MVFit beyond the prespecified +0.030 margin.
- The same shared generator supports text-only and silhouette-fitted modes; silhouette fitting changes  $\theta$  only.

- SPPA descriptors can be consumed by an optional Unreal backend that coexists with curated assets on the tested packaged/replay paths.

Not supported:

- Real UAV 3D reconstruction accuracy or measured flight telemetry.
- Universal class-agnostic open-set geometry from arbitrary words.
- Operator benefit, safety improvement, or dense VR readiness.
- Image-to-3D SOTA ranking against neural generators.

## 14 Conclusion

Family-conditioned SPPA-MVFit shows that a reviewed semantic part graph is a measurable occupancy prior for lightweight multiview proxy fitting. Under a preregistered equal-budget comparison, SPPA-MVFit improves clean 3D occupancy over Generic-MVFit by 0.190 absolute IoU (95% CI [0.181, 0.199]) on 240 held-out synthetic actors, with positive CSG-ID and implicit-OOD strata and millisecond-scale local cost. The deployment-facing SPPA contract remains a separate engineering layer for compiling imperfect semantic telemetry into updateable Unreal actors when curated assets are missing or mismatched.

The honest next steps are external public data with metric ground truth, detector-derived silhouettes, and—only if claimed—operator studies. Until then, SPPA-MVFit should be judged as synthetic confirmatory evidence for a family-conditioned knowledge representation in lightweight proxy fitting, not as a flight system or universal reconstructor.

## Data and Code Availability

The sealed SPPA-MVFit reproducibility package, protocol amendments, NIST-bound seed manifest hashes, and confirmatory summaries are included under `reproducibility/sppa_mvfit/`. Large Unreal artifacts and historical engineering logs remain supporting material and are not required to recompute the primary H1 tables.

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